

1) ESP32 CODE :-

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#include <WiFi.h>
#include <FirebaseESP32.h>
#include <BLEDevice.h>
#include <BLEScan.h>
#include <time.h>

// Firebase Configuration
#define FIREBASE_HOST "emprolltracking-default-rtdb.firebaseio.com"
#define FIREBASE_SECRET "tEsqKC0bUzeeyBQ2HCfxFZeynQoUyhX6eEXVu3gT"

// WiFi Credentials
#define WIFI_SSID "BLEPROJECT"
#define WIFI_PASSWORD "Anmol@<"

// BLE Configuration
#define SCAN_TIME 3 // Seconds per scan
#define EXIT_TIMEOUT 30 // 30 seconds without signal = exit
#define DEBOUNCE_TIME 10 // 10 seconds debounce to avoid multiple logs

FirebaseData fbdo;
FirebaseAuth auth;
FirebaseConfig config;

struct Employee {
    const char* beaconName;
    const char* empID;
    bool isActive;
    unsigned long lastSeen;
    unsigned long entryTime;
    unsigned long dailyTotal;
};

Employee employees[] = {
    {"Emp1Anmol@comp&iot", "emp1", false, 0, 0, 0},
    {"Emp2Manthan@comp&iot", "emp2", false, 0, 0, 0}
};

BLEScan* pBLEScan;
unsigned long lastReset = 0;

void setup() {
    Serial.begin(115200);

    // Initialize WiFi
    WiFi.begin(WIFI_SSID, WIFI_PASSWORD);
    Serial.print("Connecting to WiFi");
    while (WiFi.status() != WL_CONNECTED) {
        Serial.print(".");
        delay(500);
    }
    Serial.println("\nConnected: " + WiFi.localIP().toString());

    // Initialize Firebase
    config.host = FIREBASE_HOST;
    config.signer.tokens.legacy_token = FIREBASE_SECRET;
    Firebase.begin(&config, &auth);
    Firebase.reconnectWiFi(true);
```